European Journal of Sport Science

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/tejs20

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To cite this article: Gustavo F. Pedrosa, Fernando V. Lima, Brad J. Schoenfeld, Lucas T. Lacerda, Marina G. Simões, Mariano R. Pereira, Rodrigo C.R. Diniz & Mauro H. Chagas (2021): Partial range of motion training elicits favorable improvements in muscular adaptations when carried out at long muscle lengths, European Journal of Sport Science, DOI: 10.1080/17461391.2021.1927199

To link to this article: https://doi.org/10.1080/17461391.2021.1927199

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EUROPEAN JOURNAL OF SPORT SCIENCE
https://doi.org/10.1080/17461391.2021.1927199

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Partial range of motion training elicits favorable improvements in muscular adaptations when carried out at long muscle lengths

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ABSTRACT

The study compared changes in strength and regional muscle hypertrophy between different ranges of motion (ROM) in the knee extension exercise. Forty-five untrained women were randomized to either a control group or to perform the exercise in one of the following 4 groups (0°=extended knee): Full ROM ($FULL_{ROM}$: 100°–30° of knee flexion); Initial Partial ROM ($INITIAL_{ROM}$: 100°–65°); Final Partial ROM ($FINAL_{ROM}$: 65°–30°); Varied ROM (VAR_{ROM}: daily alternation between the ROM of $INITIAL_{ROM}$ and $FINAL_{ROM}$). Pre- and post-training assessments included one repetition maximum (1RM) testing in the ROM corresponding to the initial, final and full ROM, and measurement of cross-sectional areas of the rectus femoris and vastus lateralis muscles at 40%, 50%, 60% and 70% of femur length in regard to regional muscle hypertrophy. Results showed that the $INITIAL_{ROM}$ group presented a greater relative increase than all groups at 70%, and at 50% and 60% the increases were greater than $FINAL_{ROM}$, $FULL_{ROM}$, and non-training control (CON) groups. Moreover, $FINAL_{ROM}$ group presented similar changes compared to the CON group at 60% and 70%. In regard to 1RM, $FINAL_{ROM}$ and $INITIAL_{ROM}$ groups presented greater relative increases at the ROM trained, and no group showed greater increases than VAR_{ROM} or $INITIAL_{ROM}$, regardless the ROM tested. In conclusion, partial ROM training in the initial phase of the knee extension exercise promoted greater relative hypertrophy in certain muscle regions than training in other ROM configurations, and no group promoted a greater 1RM increase than VAR_{ROM} group, which showed similar 1RM increases in the different ROMs tested.

KEYWORDS
Muscle hypertrophy; resistance exercise; partial range of motion; excursion

Introduction

Regional muscle hypertrophy is an adaptive response to resistance training that occurs when the size of given muscle region increases to a greater extent than other muscle regions (Zabaleta-Korta, Fernández-Peña, & Santos-Concejero, 2020). It has been speculated that training in different joint ranges of motion (ROM) may indeed promote such non-uniform adaptations (Newmire & Willoughby, 2018; Zabaleta-Korta et al., 2020). Studies by Bloomquist et al. (2013) and McMahon, Morse, Burden, Winwood, and Onambélé (2014a) demonstrated that training in a full ROM ($FULL_{ROM}$) elicited greater muscle hypertrophy at distal muscle regions than training in the final partial ROM ($FINAL_{ROM}$: i.e. final half of the angles of a $FULL_{ROM}$, taking the concentric action as reference). Of note, both studies observed similar hypertrophic responses between conditions at the proximal muscle regions. These findings suggest that training in a $FINAL_{ROM}$ preferentially induces greater muscle hypertrophy at proximal regions than in other regions, while training with $FULL_{ROM}$ equally hypertrophies the muscle across the regions. However, neither study compared the hypertrophy responses between muscle regions within each training condition, nor did they compare the hypertrophy of different muscle regions between different muscles. These inter- and intramuscular analyses would provide greater insight as to whether, and the extent to which, ROM may influence a regional hypertrophic response.

The apparent hypertrophic superiority of $FULL_{ROM}$ training may be specific to contractions at angles in which the muscle is elongated (Alegre, Ferri-Morales, Rodriguez-Casares, & Aguado, 2014; Maeo et al., 2020; Noorkõiv, Nosaka, & Blazevich, 2014). Previous research

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Supplemental data for this article can be accessed https://doi.org/10.1080/17461391.2021.1927199.
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reports greater metabolic stress, IGF-1 release, and activation of proteins Akt/mTOR and p70S6 K after contractions at longer vs shorter muscle lengths (McMahon, Morse, Burden, Winwood, & Onambélé, 2014b; Rindom, Kristensen, Overgaard, Vissing, & de Paoli, 2019; Russ, 2008), all of which have been associated with muscle hypertrophy. This raises the possibility that training exclusively in the initial half of the angles of a $FULL_{ROM}$ ($INITIAL_{ROM}$, taking the concentric action as reference), where the muscle is in a lengthened state, may promote a hypertrophic response even greater than training throughout a $FULL_{ROM}$.

Recently, researchers have speculated that training through a variety of ROMs (VAR_{ROM}) may enhance muscle growth (Schoenfeld & Grgic, 2020). One such variation strategy could be to alternate between training with $INITIAL_{ROM}$ and $FINAL_{ROM}$ from session to session. This undulating model resembles the muscle length variations that occur during $FULL_{ROM}$ training, but allows the ability to focus on the individual components of the ROM which conceivably may enhance muscular adaptations. However, this hypothesis has yet to be tested.

ROM also has potential implications for the strength-related response to resistance training. Consistent with the principle of specificity (McCafferty & Horvath, 1977), current evidence indicates greater strength increases are achieved at the ROM trained (Martínez-Cava et al., 2019; Pallarés, Cava, Courel-Ibáñez, González-Badillo, & Morán-Navarro, 2020; Weiss, Fry, Wood, Relyea, & Melton, 2000). For example, Bloomquist et al. (2013) found that training exclusively in the $FINAL_{ROM}$ resulted in a greater strength increase in the one maximum repetition (1RM) test performed at the ROM trained. Thus, the question remains as to whether strength gains also would be specific to the ROM tested when training is performed in the $INITIAL_{ROM}$. Moreover, it is undetermined as to what effect using a VAR_{ROM} would have on specific strength changes over time.

To address the aforementioned gaps in the literature, the present study aimed to compare the effects of training with $INITIAL_{ROM}$, $FINAL_{ROM}$, $FULL_{ROM}$, and VAR_{ROM} on regional muscle hypertrophy and the 1RM strength response performed in different ROMs. In addition, we aimed to assess the relationship between the strength increases in different ROMs with the muscle growth in different muscle regions.

Methods

Overview

Participants equally comprised 4 experimental groups and 1 control group (CON) over a 12-week study period. The experimental groups trained in one of 4 different ROM configurations in the knee extension machine ($INITIAL_{ROM}$:, $FINAL_{ROM}$, $FULL_{ROM}$, and VAR_{ROM}) for 36 training sessions separated by 48–72 h (Figure 1).

To assess regional changes in hypertrophy of the rectus femoris (RF) and vastus lateralis (VL) muscles, pre- and post-training measures of muscle cross-sectional area (CSA) were obtained via b-mode ultrasound at 4 longitudinal regions along the femur (40%, 50%, 60%, and 70% of femur length). Changes in dynamic muscular strength were assessed via 3 different bilateral 1RM tests in the knee extension corresponding to the $INITIAL_{ROM}$, $FINAL_{ROM}$, and $FULL_{ROM}$.

Subjects

Forty-five non-trained women with an age range of 18–30 years old (mean ± SD: age = 22.7 ± 2.8 years; weight = 61.5 ± 9.0 kg; height = 1.61 ± 0.01 m; body fat percentage = 25.9 ± 5.1%) participated in the study. The local ethics committee approved this study, which complied with international standards. In addition, each subject was instructed not to perform any physical activity on the testing session's days.

Procedures

Pre-training session 1

Height, body mass, and fat percentage (skinfold thickness) measurements were performed. Thereafter, ultrasound images were obtained to measure CSA of the RF and VL muscles. Initially, participants remained still on a stretcher in the supine position for 15 min. During this period, the anterior regions of the right thigh were marked to identify the points of image acquisition. For the marking procedure, femur length was measured as the distance between the major trochanter and lateral epicondyle. From the proximal extremity, mark points were made on the volunteer's thigh at 40%, 50%, 60%, and 70% of femur length using a tape measure. A laser device was used to mark the other points and verify their axial alignment. This device was placed over the volunteer's thigh, and the lasers indicated points axially aligned across the thigh (check supplementary material). The procedures used to acquire images in the pre-training were the same for the post-training, which was completed between 72–96 h after the last training session.

A b-mode ultrasound device (MindRay DC-7, Shenzhen, China) with a 4-cm linear transducer was used in extended-field-of-view mode to assess muscle CSA. The equipment was configured following Lacerda et al. (2021) and adjusted for each individual participant and muscle region to produce the clearest possible images. The same trained evaluator performed the acquisition

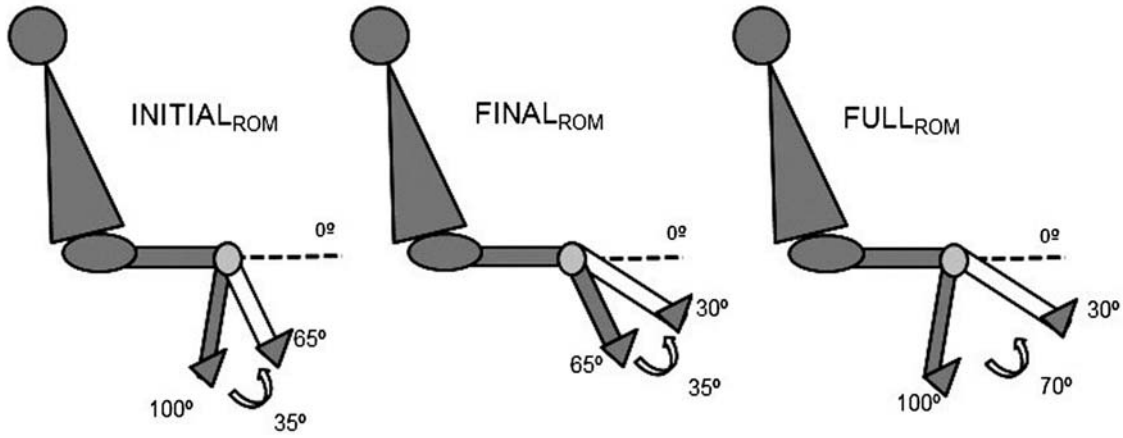

Figure 1. Training groups. VAR_{ROM} group daily alternated the ROM between $INITIAL_{ROM}$ and $FINAL_{ROM}$.

of 2 images at each percentage of femur length (40%, 50%, 60%, and 70%). The probe was placed transversely in parallel to the intercondylar line using a coupled guide on the volunteer's thigh. A set of sixteen images per volunteer were obtained for RF and VL CSA analysis (8 pre-training + 8 post-training). After obtaining the images, the CSA of each muscle scan was manually estimated by a blinded examiner using specific software (OsiriX MD 6.0, Bernex, Switzerland). The RF and VL CSA regional values were determined by calculating the mean values of the 2 images acquired at each percentage of femur length (Figure 2).

Reliability of CSA measurements was assessed by analysis of the intraclass correlation coefficient ($ICC_{3,1}$) for both intra- and inter-evaluator. For intra-evaluator, the 2 CSA measures of each region of the RF and VL demonstrated $ICC_{3,1}$ values between 0.98–0.99. For inter-evaluator reliability, an invited radiology technician measured the CSA of 1/3 of all images. The inter-evaluator reliability showed $ICC_{3,1}$ values ranging from 0.94–0.99 along the 4 muscle regions of the two muscles.

We also measured muscle thickness at 50% of length of the RF muscle, determined as the distance between the deeper and upper RF muscle aponeurosis (Miyatani, Kanehisa, Kuno, Nishijima, & Fukunaga, 2002). The muscle thickness measurement was ranked in order of size and used as a criterion for the balanced randomization of the participants into the groups. We opted to use muscle thickness for randomizing the participants because of its ease of measurement compared to the CSA and given the need to complete the randomization process prior to the ensuing pre-training session.

Pre-training sessions 2, 3, 4, and 5

During pre-training session 2, participants were familiarized with the bilateral 1RM tests corresponding to the 3 ROM tests ($INITIAL_{ROM}$, $FINAL_{ROM}$, and $FULL_{ROM}$ group). First, the participants were positioned in the knee extension machine with 110° of hip flexion (trunk and thigh) and the medial malleolus of the tibia positioned 2 cm below the machine pad. To minimize accessory movements, participants were restrained in the machine by a four-point seat belt. In addition, a metallic structure was placed in front of the machine to serve as a reference (mechanical stop) to the desired knee extension angle (65° or 30° of knee flexion) (see supplementary material). Also, a potentiometer, coupled to the rotational axis of the mechanical arm of the knee extension machine, provided real-time biofeedback on a computer screen as to the ROM excursed by each volunteer. The use of a potentiometer to analyze the ROMs excursed during tests and training were described in previous studies of our group (Diniz et al., 2020; Lacerda et al., 2021).

The familiarization to 1RM tests was determined within a maximum of 6 attempts, with 3-minute rest periods provided between attempts. A 30-minute rest period was given between the tests (order balanced between the participants) for familiarization.

In the third, fourth, and fifth sessions at pre-training, a sole 1RM test was performed in each session, which followed the procedures, and order of the ROM tested in familiarization. The data of these 3 sessions were used for statistical analysis. The range of $ICC_{3,1}$ values between the performance on familiarizations and tests were 0.94–0.97.

Training

After the initial testing period, participants carried out a 12-week training programme using their designated ROM. The experimental protocol consisted of 3–6 sets of 7 repetitions (each repetition carried out with 2-

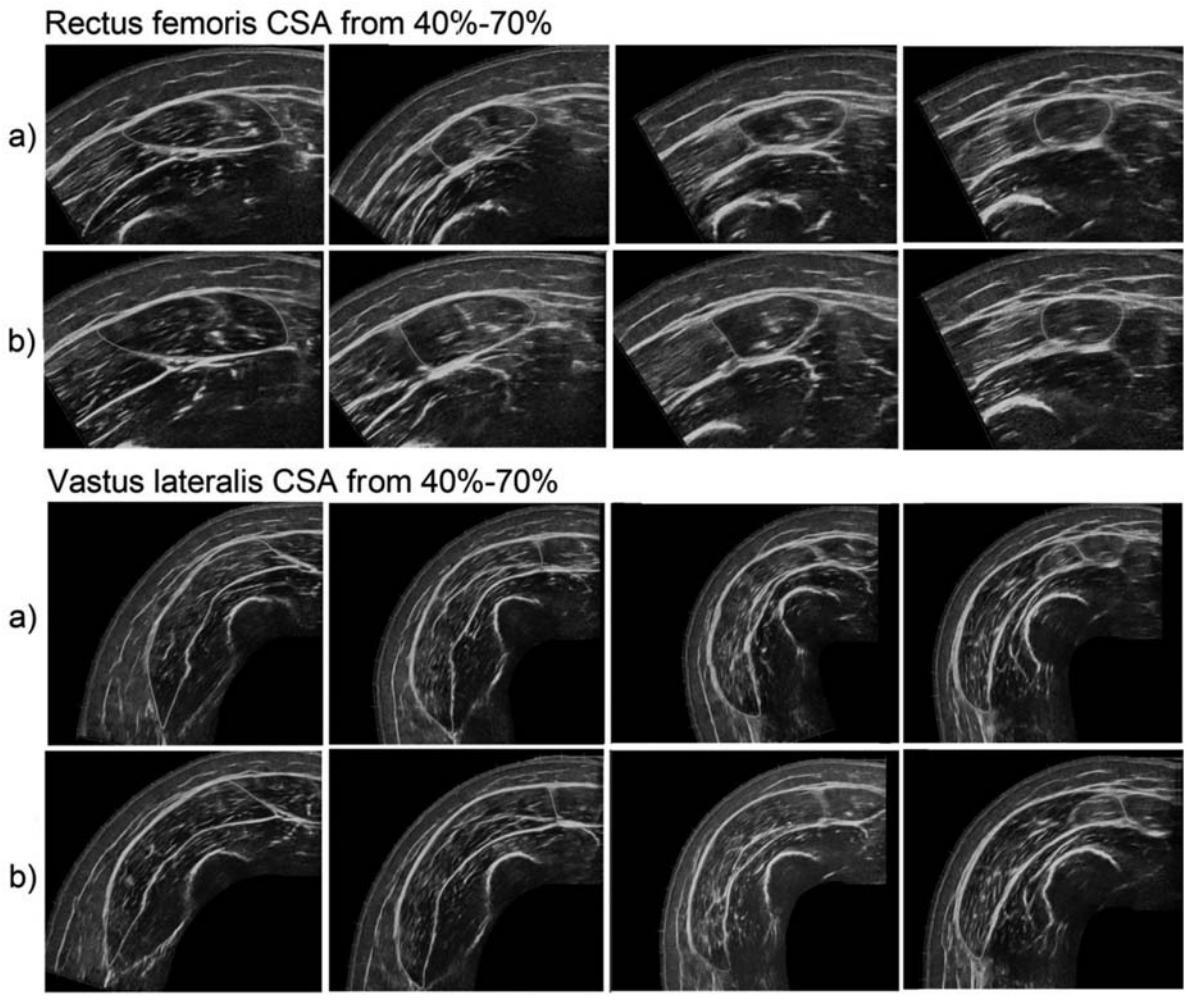

Figure 2. Example of extended-field-of-view of four regions of rectus femoris and vastus lateralis muscles from the same volunteer**.** CSA = Cross-sectional area. a) = pre-training. b) = post-training.

second concentric and 2-second eccentric actions, with the tempo controlled by a metronome) at 60% 1RM with 3-minute rest periods between sets. The 1RM percentage was obtained specific to the ROM performed in a given group. All training groups started the training period performing 3 sets. This training load configuration was based on a pilot study, which indicated that most individuals (similar to the present study) were unable to perform more than 7 repetitions across 3 sets (intensity at 60% of 1RM, specific to each ROM with 2-second concentric and 2-second eccentric actions, and a 180-second rest interval between sets). During the 3rd to 5th week of training, the women performed 4 sets. Between the 6th and 8th week 5 sets were performed, and between the 9th and 12th week of training the participants performed 6 sets. The mechanical stops used during the 1RM tests to delimitate the ROM desired and the potentiometer were also used in the training period. The data provided by the potentiometer allowed the participants to assess the duration and ROM data of each muscle action on a laptop screen during all training sessions and tests. To verify that groups trained according to their prescribed muscle action duration and ROM, we randomly analyzed 1/6 of all training sessions. No statistical difference was observed in the mean muscle action duration among groups (one-way ANOVA – $p>0.05$). Also, no difference was observed among groups with similar start or end angles (one-way ANOVA – $p>0.05$).

Every 2 weeks of training, the participants performed a 1RM test in the ROM trained to maintain the intensity of load programmed throughout the 12 weeks of training. These tests occurred before the start of the training session; a 10-minute rest period separated the completion of 1RM testing and the onset of the training session. Assessment of rating of perceived exertion found that this protocol elicited a mean rating of 15 on the Borg category scale (range of 6–20) across all sets, corresponding to a "hard" level of perceived effort; the final set tended to be rated as "very hard,"

indicating that workouts were sufficiently challenging for all groups.

Post-training sessions 1, 2, and 3

In the first post-training session, which occurred between 72–96 h after the final training session, all participants were submitted to ultrasound assessment of the same muscles, regions, and procedures as conducted in pre-training. Afterward, the participants performed the first 1RM test following the order and procedures determined at pre-training. In the second and third sessions of post-training (48 h apart between sessions) the other two 1RM tests were performed with the same order and procedures of the pre-training sessions.

Statistical analyses

Statistical analysis was performed with Sisvar version 5.7 software. We carried out analyses of variance (ANOVA) to test for differences in absolute baseline values for all variables analyzed, with no differences identified between groups. To verify the change caused by training protocols performed by ROM manipulation, we transformed the CSA and 1RM test performance data into relative responses ((Post – Pre)/Pre x 100). The normality and homogeneity of variances were verified using Shapiro–Wilk and Levene's tests, respectively. Eta-squared (η^2) was calculated for the main significant effect (Fritz, Morris, & Richler, 2012), with the following interpretation: small<0.06; medium=0.06–0.14; large>0.14 (Cohen, 1988). To compare the CSA relative responses, a three-way ANOVA was performed (Group x Muscle x Region). Moreover, the 1RM relative responses were compared with a two-way ANOVA (Group x ROM). To verify whether the results of the 1RM tests were different among groups and ROMs tested at pre-training, we employed a two-way ANOVA (Group x ROM). In the case of statistical differences, the Scott-Knott post hoc test was employed to determine which groups differed (Scott & Knott, 1974). Pearson correlation coefficient was used to verify the relationship between the regions CSA relative changes and 1RM tests performance relative changes. Data are presented as mean ± SD. Probability was set at $\alpha \leq 0.05$ for statistical significance for all responses.

Results

Cross-sectional area

The three-way ANOVA showed a significant interaction effect between the factors Group x Muscle (p=0.0154; η^2=0.04). For the RF, $INITIAL_{ROM}$ and VAR_{ROM} groups presented a greater change than the other groups, and the $FULL_{ROM}$ group presented a greater change than $FINAL_{ROM}$ and CON groups, which showed similar changes. For the VL, the $INITIAL_{ROM}$, $FULL_{ROM}$, and VAR_{ROM} groups presented similar responses, which were greater than $FINAL_{ROM}$ and CON groups; the $FINAL_{ROM}$ group presented a greater increase than the CON group. Furthermore, $INITIAL_{ROM}$ and VAR_{ROM} groups presented a greater increase for the RF than the VL. For the other groups, both muscles presented similar changes.

A significant interaction effect was found between the factors Group x Region (p=0.0349; η^2=0.01). The post hoc test revealed similar changes between the training groups, which were all greater than the CON group at 40% of femur length. $INITIAL_{ROM}$ and VAR_{ROM} groups presented similar changes and greater than the other groups at 50%, whereas $FULL_{ROM}$ and $FINAL_{ROM}$ groups presented similar increases, which were greater than the CON group. $INITIAL_{ROM}$ and VAR_{ROM} groups presented similar changes and greater than the other groups at 60%, followed by the $FULL_{ROM}$. Moreover, $FINAL_{ROM}$ and CON groups showed similar responses. $INITIAL_{ROM}$ group presented greater responses than other groups at 70%. Additionally, $FULL_{ROM}$ and VAR_{ROM} groups showed similar increases and greater than $FINAL_{ROM}$ and CON groups, which showed similar relative responses. The CSA changes among the 4 muscle regions were similar for the $INITIAL_{ROM}$, $FULL_{ROM}$, VAR_{ROM} and CON groups. However, for the $FINAL_{ROM}$ group, the CSA changes at 40%, 50% and 60% were greater than at 70% (Figure 3).

1RM Tests

Two-way ANOVA found an interaction between the factors Group x ROM (p=0.0394; η^2=0.14). The post hoc test revealed that in 1RM testing performed at the initial ROM, $INITIAL_{ROM}$ and VAR_{ROM} groups showed greater changes than the other groups, and $FINAL_{ROM}$ and $FULL_{ROM}$ groups showed similar changes, which were greater than CON group. For the 1RM test performed at the final ROM, the training groups showed similar increases, all of which were greater than the CON group. In the 1RM test performed at the full ROM, $INITIAL_{ROM}$, $FULL_{ROM}$, and VAR_{ROM} groups presented similar relative increases, which were greater than $FINAL_{ROM}$ and CON groups; $FINAL_{ROM}$ showed greater changes than CON group. Additionally, $INITIAL_{ROM}$ and $FINAL_{ROM}$ groups presented a greater increase in

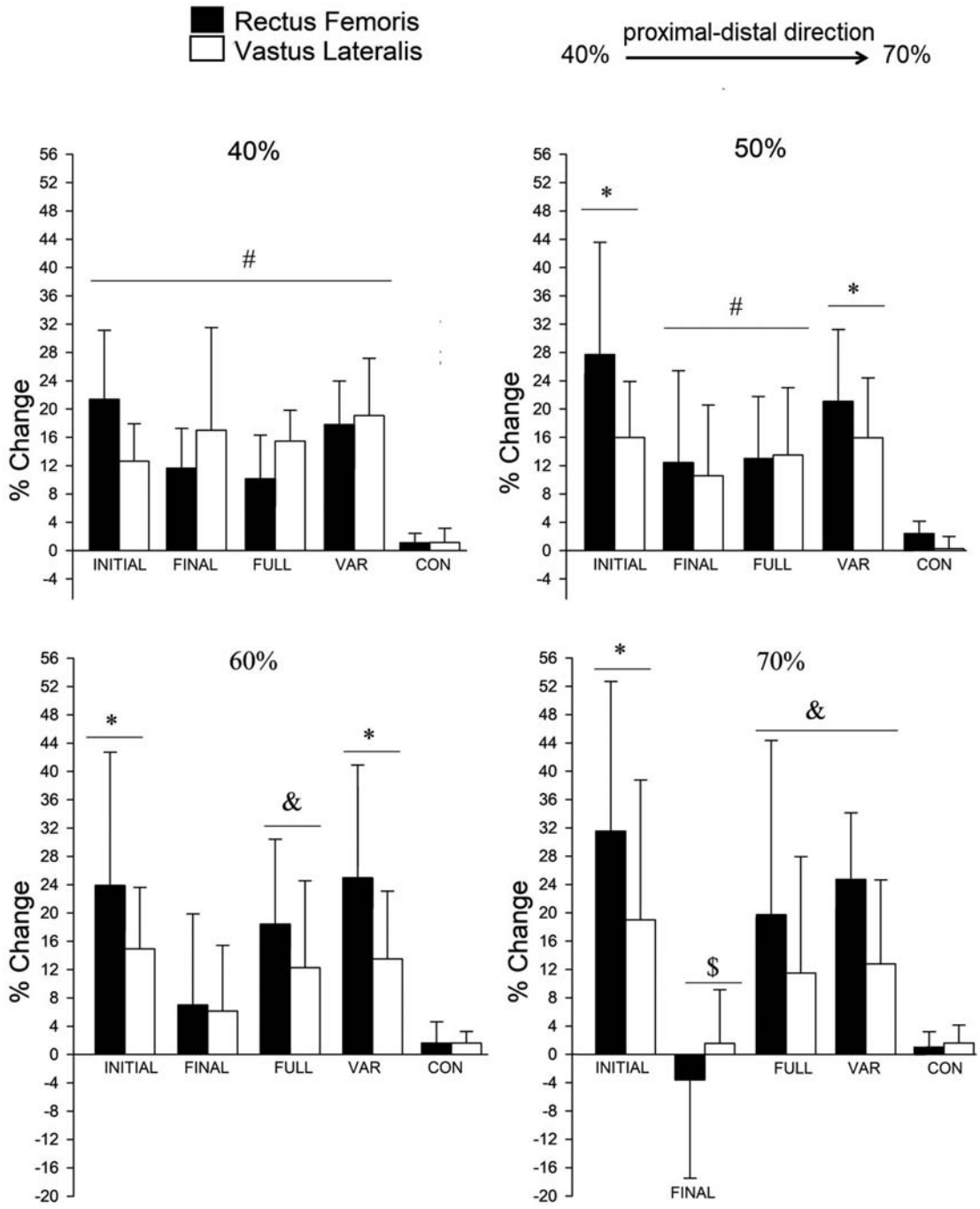

Figure 3. The relative increase of cross-sectional area at 40%, 50%, 60%, and 70% of femur length. INITIAL = $INITIAL_{ROM}$; FINAL = $FINAL_{ROM}$; FULL = $FULL_{ROM}$; VAR = VAR_{ROM}. *Greater increase than the other groups; &Greater increase than the $FINAL_{ROM}$ and CON groups; #Greater increase than CON group; $Different than at other regions for the same group.

the 1RM test at the respective ROM trained, and the other groups did not show differences between the tests (Figure 4).

From a loading standpoint, two-way ANOVA only found a main effect for ROM whereby a similar amount of weight was lifted at the $INITIAL_{ROM}$ and $FINAL_{ROM}$, and at these two ROMs, a greater amount of weight was lifted than at the full ROM (p=0.034; η^2=0.05).

Pearson correlation coefficient

For the $INITIAL_{ROM}$ group, the Pearson correlation coefficient indicated a significant, and positive association between relative CSA changes of the VL at 60% and 70% and the 1RM tests at the initial and full ROM (r=0.73–0.80; p=0.01–0.02). For the $FINAL_{ROM}$ group, a positive and significant relationship was observed between the changes of CSA of VL at

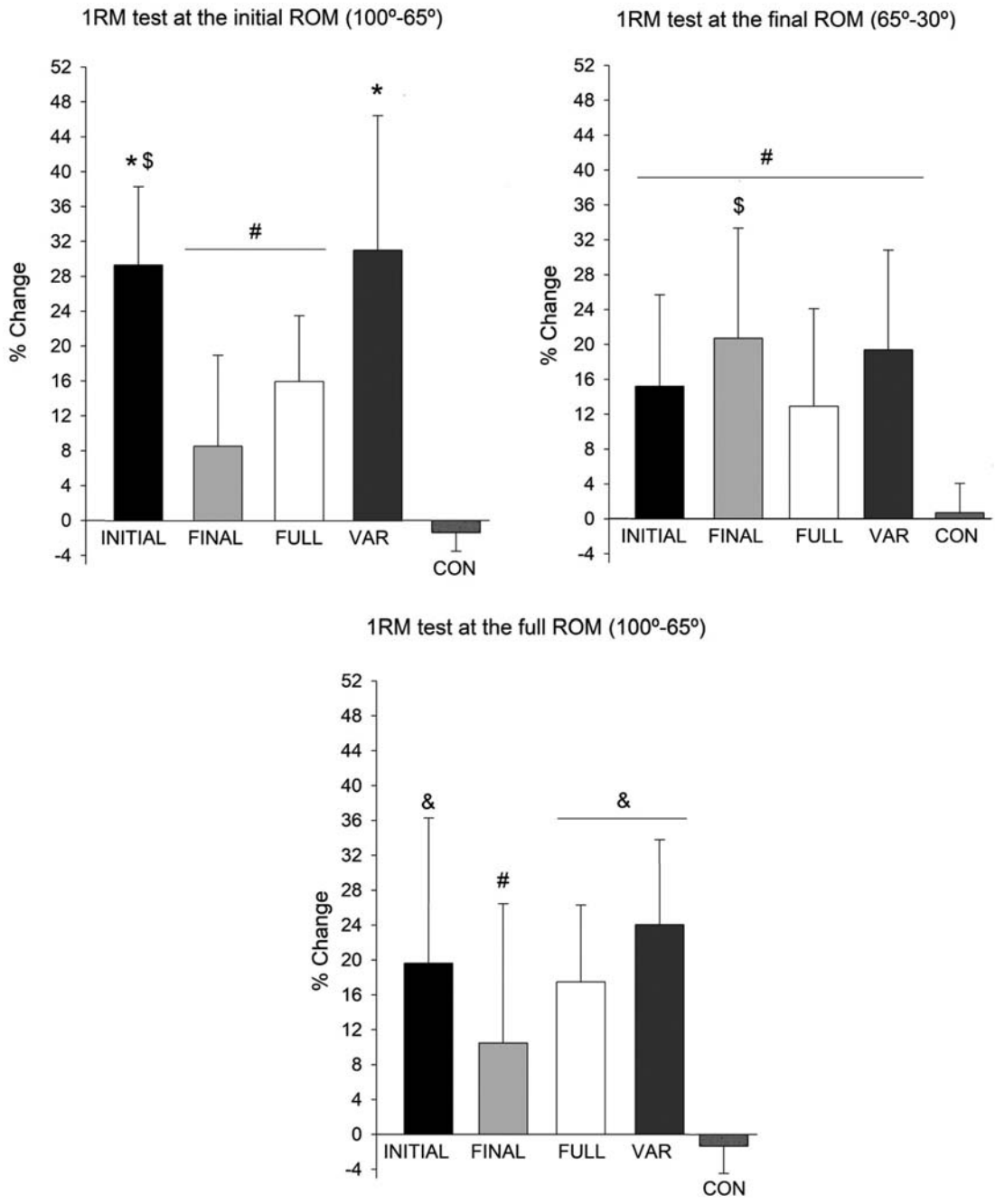

Figure 4. 1RM tests in different ROM. INITIAL = $INITIAL_{ROM}$; FINAL = $FINAL_{ROM}$; FULL = $FULL_{ROM}$; VAR = VAR_{ROM}. *Greater than the other groups at the respective ROM; &greater than groups $FINAL_{ROM}$ and CON at the respective ROM; # only greater than CON Group at the respective ROM. $ different than the other ROM for the same group. L = lower values. U = upper values.

40%, 50% and 60% with the 1RM tests at the final and full ROM (r=0.72–0.98; p<0.001–0.020), and at 70% with the 1RM test at full ROM (r=0.70; p=0.037). For the VAR_{ROM} group, the CSA increases of the RF at 60% and 70% presented a significant correlation with the 1RM test increase at the initial ROM (r=0.68–0.76; p=0.044–0.018). For the $FULL_{ROM}$ and CON groups, no significant correlation was found.

Discussion

Muscle hypertrophy

The $INITIAL_{ROM}$, $FULL_{ROM}$, and VAR_{ROM} groups elicited similar CSA changes across the muscle regions. Although these groups excursed the initial ROM (100°–35° of knee flexion), only the $INITIAL_{ROM}$ group exclusively excursed this ROM. Therefore, excursing a full ROM may not be causal for inducing homogenous muscle growth across

muscle regions, as previously hypothesized (Helms, Fitschen, Aragon, Cronin, & Schoenfeld, 2014), but rather due to training at a long muscle length in the initial ROM. McMahon et al. (2014b) showed that after 8 weeks of training with the knee extension, the $INITIAL_{ROM}$ group presented a greater VL change at the distal region than the $FINAL_{ROM}$ group. This result casts doubt as to whether the $INITIAL_{ROM}$ training does in fact promote a greater distal muscle change compared to other regions and raises the alternative possibility that $FINAL_{ROM}$ training elicits less change in the distal region than at other regions. Overall, the present results suggest that excursing the initial ROM during knee extension exercise elicits similar growth across the muscle regions, while training with final ROM may not significantly hypertrophy the distal muscle regions, at least in the muscles analyzed. However, the present study analyzed only four muscle regions of two quadriceps femoris muscles, thus preventing the extrapolation of these results to other regions and muscles of the quadriceps femoris.

When considering regional hypertrophic changes between groups, the $INITIAL_{ROM}$ group achieved greater relative increases in CSA at 50% and 60% compared to $FINAL_{ROM}$, $FULL_{ROM}$, and CON groups, and at 70% adaptations were greater than all groups. Given that the protocol for $FINAL_{ROM}$, $FULL_{ROM}$, and VAR_{ROM} groups included at least some training at shorter muscle lengths, it is possible a concentrated focus on training at a longer muscle length by the $INITIAL_{ROM}$ group was more advantageous for muscle development. Indeed, previous research shows a hypertrophic superiority to training at longer versus shorter muscle lengths (Alegre et al., 2014; McMahon et al., 2014b; Noorkõiv, Nosaka, & Blazevich, 2015). Our results expand upon these findings to indicate that excursing shorter muscle lengths may be detrimental to hypertrophic changes even when some of the training is carried out at longer muscle lengths. Intriguingly, analysis of the 1RM training loads showed that both partial groups ($INITIAL_{ROM}$ and $FINAL_{ROM}$) employed heavier loads than the $FULL_{ROM}$ group, yet the $INITIAL_{ROM}$ group achieved greater increases in CSA than the other two groups. Since the magnitude of load was similar for the $INITIAL_{ROM}$ and $FINAL_{ROM}$ groups, this would seem to indicate that mechanical stress alone was not responsible for the observed differences; rather, muscular development appears to be driven via an interaction between mechanical stress and muscle length, the mechanisms of which are not clear.

It is worth noting that the higher relative increases in CSA between groups occurred at the distal muscle region (~30% for RF and ~18% for VL) for the $INITIAL_{ROM}$ group, which is in line with prior studies that investigated muscle hypertrophy in different regions (proximal, middle and distal) via ultrasound imaging (McMahon et al., 2014a; 2014b). McMahon et al. (2014b) found that knee extension training in the initial ROM promoted a ~50% CSA increase of the distal VL (75% of femur length). In another study by the same group, McMahon et al. (2014a) reported a 40.1% CSA increase of the distal VL (75% of femur length) after training in a full ROM using a variety of different knee extension exercises. In both studies, these hypertrophic increases were relatively greater than that achieved proximally, and were superior at longer versus shorter muscle lengths.

Considering the hypertrophic differences between the $INITIAL_{ROM}$ and $FULL_{ROM}$ groups, it seems the combination of training with a longer time under tension at a longer muscle length with higher loads (ascertained by greater pre- 1RM test values at the partial ROMs than at full ROM) for the $INITIAL_{ROM}$ group provided a greater stimulus for hypertrophy than training in full ROM, which likely occurred with greater mechanical work (greater angular displacement). Previous studies indicate that metabolic stress (Fouré, Ogier, Guye, Gondin, & Bendahan, 2020; Kooistra, Blaauboer, Born, de Ruiter, & de Haan, 2005; Ng, Agre, Hanson, Harrington, & Nagle, 1994) and activation of the proteins Akt/mTOR and p70S6 K (Rindom et al., 2019; Russ, 2008) can occur to a greater magnitude when performing contractions at longer versus shorter muscle lengths. This evidence suggests a heightened anabolic milieu at longer compared to shorter muscle length contractions, which conceivably may help to explain the discrepancy of results, as the $FULL_{ROM}$ group trained in both a shorter and longer muscle length, while the $INITIAL_{ROM}$ group exclusively trained at a longer muscle length. However, the mechanisms underpinning regional hypertrophy due to contractions at longer and shorter muscle lengths are still poorly characterized and warrant further research.

$INITIAL_{ROM}$ and VAR_{ROM} groups showed similar regional hypertrophic responses at 40%, 50%, and 60%. We speculated that hypertrophy would be similar between VAR_{ROM} and $FULL_{ROM}$, which excursed of the spectrum of initial and final ROMs during each repetition. Contrary to this hypothesis, regional muscular gains were generally greater in VAR_{ROM} than $FULL_{ROM}$. A possible explanation could be related to the fact that the VAR_{ROM} group trained with a higher load than the $FULL_{ROM}$ group at a longer muscle length. Conceivably, the greater mechanical stress imposed under the condition of a stretched muscle may therefore promote a synergistic anabolic stimulus.

Regarding the individual muscles investigated, the RF and VL presented similar relative increases in the $FINAL_{ROM}$ and $FULL_{ROM}$ groups. Alternatively, the RF muscle increased to a greater magnitude than the VL in the $INITIAL_{ROM}$ and VAR_{ROM} groups. These results suggest that training at longer muscle lengths may promote greater hypertrophy for the RF compared to the VL. This result runs counter to that of Noorkõiv et al. (2014), who reported similar increases in muscle volume between the RF and VL when training isometrically at a longer muscle length. Further investigation is needed to clarify whether training at long muscle lengths does in fact promote a greater hypertrophy response for the RF than for the VL.

In regard to the regional muscle hypertrophy within groups, The $FINAL_{ROM}$ group presented greater regional muscle growth at more proximal regions (40% and 50%) than at more distal regions (60% and 70%). These results may help to clarify findings of previous research, in which the $FINAL_{ROM}$ groups showed a lower hypertrophic response at distal regions compared to $FULL_{ROM}$ (Bloomquist et al., 2013; McMahon et al., 2014a) and $INITIAL_{ROM}$ groups (McMahon et al., 2014b). Bloomquist et al. (2013), despite not having performed comparisons among muscle regions within each training group ($FINAL_{ROM}$ x $FULL_{ROM}$), performed paired t-tests to compare longitudinal CSA of pre x post values in 6 regions across the anterior part of the thigh after squatting training. For the $FINAL_{ROM}$ group, the results showed that only the two most proximal muscle regions increased after training. These data reinforce our findings, which suggest that training at shorter muscle lengths may not promote significant muscle hypertrophy at more distal regions in the RF and VL muscles. On the other hand, training at a longer muscle length seems to promote a homogeneous hypertrophic response across the muscle regions, as occurred for the $INITIAL_{ROM}$ $FULL_{ROM}$ and $VAR_{RO.}$ However, our data are limited to four regions of two quadriceps femoris muscles. Future studies should seek to investigate more muscle regions within a given muscle as well as muscles other than the quadriceps femoris to enhance our understanding of the effect of ROM on the regional muscle hypertrophy response.

Dynamic strength

Based on the principle of specificity (McCafferty & Horvath, 1977), we expected that greater increases in dynamic strength would occur in the ROM trained. This hypothesis was partially confirmed, as the $FINAL_{ROM}$ and $INITIAL_{ROM}$ groups achieved a greater strength increase in the respective ROM tested; the result of the $FINAL_{ROM}$ group is consistent with those reported in previous studies (Bloomquist et al., 2013; Martínez-Cava et al., 2019; Rhea et al., 2016). Although no study to date has investigated strength changes in the $INITIAL_{ROM}$, the results of study that investigated isometric torque production across angles after training with partial ROM corroborate the present findings, due to increases in isometric torque occurred near or at the training angles (Graves, Pollock, Jones, Colvin, & Leggett, 1989, 1992). Additionally, the strength gains from training at a longer muscle length may also occur further along the training angle (Noorkõiv et al., 2015; Thepaut-Mathieu, Van Hoecke, & Maton, 1988). These results support the present findings, as the $INITIAL_{ROM}$ group demonstrated strength increases similar to the groups that showed a greater strength increase in the 1RM test at the final and full ROM.

Previous research suggests that neural adaptations help to explain joint-angle-specific strength increases, with increasing muscle activation observed at or near the angles trained (Alegre et al., 2014; Noorkõiv et al., 2014; Thepaut-Mathieu et al., 1988). Nevertheless, regional muscle hypertrophy may also play a role in this phenomenon (Narici, Roi, Landoni, Minetti, & Cerretelli, 1989; Noorkõiv et al., 2014). Noorkõiv et al. (2014) found positive and significant correlations between the relative changes of regional CSA and the torque at angles near the angle trained during isometric knee extension exercise. However, these results account only for the group that trained isometrically at a longer muscle length, as no significant correlation was found when training at a shorter muscle length. Based on these results, Noorkõiv et al. (2014) speculated that specific regional CSA increases may be associated with increasing torque production at or near the angles trained after training in longer muscle length. However, the group that trained at a shorter muscle length in the Noorkõiv et al. (2014) did not show significant hypertrophy in any muscle analyzed after 6 weeks of training, but this group did display isometric strength increases near the training angles. The present study partly corroborates the findings of Noorkõiv et al. (2014), as $INITIAL_{ROM}$ and $FINAL_{ROM}$ groups presented positive and significant correlations between regional CSA increases and 1RM increases at the ROM trained. Collectively, the present study suggests that the musculature can remodel regionally to enhance strength production, but this variation seems to be dependent on the ROM trained.

$FULL_{ROM}$ and VAR_{ROM} were the unique training groups that presented similar 1RM test increases among the ROM tested. These results are consistent with previous studies, in which the $FULL_{ROM}$ group

showed uniform improvements in 1RM test performance at the different ROM tested (Martínez-Cava et al., 2019; Pallarés et al., 2020; Weiss et al., 2000). Since both $FULL_{ROM}$ and VAR_{ROM} groups excursed the spectrum of angles investigated (100°–30° of knee flexion), we hypothesize that the overload imposed at each joint angle may have been sufficient to promote adaptations leading to similar strength increases in all ROMs investigated, which would be in accordance with Rhea et al. (2016). However, VAR_{ROM} presented a greater 1RM increase than $FULL_{ROM}$ in the initial ROM, but not in the final or full ROM. Despite the observation that the $FULL_{ROM}$ group likely trained with greater mechanical work across the repetitions, the VAR_{ROM} group trained with a higher load across the angles trained by the $FULL_{ROM}$ group. These data suggest that knee extension training with a higher load in the initial ROM was more advantageous to strength increases in the initial ROM as opposed to training with greater mechanical work across a full ROM. These findings possibly may be explained by the hypertrophic adaptations experienced by each training group. The VAR_{ROM} group presented a greater muscle hypertrophy increase than $FULL_{ROM}$ group at 50% and 60%, and some evidence suggests that region-specific increases of muscle mass may be associated with dynamic strength increases at certain training angles (Noorkõiv et al., 2015). Thus, the greater muscle hypertrophy increase may have accounted for the VAR_{ROM} group to achieve a greater 1RM increase in the $INITIAL_{ROM}$ compared to the $FULL_{ROM}$. In support of this line of reasoning, the CSA increases at 60%, and 70% of RF in the VAR_{ROM} group significantly correlated with the 1RM test increases in the initial ROM. The implications of these findings require further investigation as our analysis is limited to four muscle regions of two quadriceps muscles.

Intriguingly, despite the fact that the VAR_{ROM} group also trained with a higher load than $FULL_{ROM}$ group in the final ROM, the 1RM increase in the final ROM was similar between these groups. It seems the impact of training with higher loads in the final and initial ROMs compared to training in a full ROM induces different strength responses; these findings require further study as to the associated morphological and neural adaptations.

VAR_{ROM} group was the only group that presented a relative increase similar to the $INITIAL_{ROM}$ group in the initial 1RM test. Moreover, the VAR_{ROM} group presented similar relative increases in dynamic strength across the employed 1RM tests. These results indicate that daily undulating training between the initial and final ROMs can promote similar increases in dynamic strength across both partial and full ROMs. Moreover, these strength gains are equivalent to the groups that trained exclusively in one of these ROMs. Further research is needed to better characterize potential mechanisms and the application of undulating training between different partial ROMs.

Conclusion

The present study indicates that training in different ROM configurations may lead to regional changes in muscle CSA and dynamic strength performance. Resistance training at the initial ROM generally showed superiority in comparison to final, full, and varied ROM training. In addition, maximum strength performance is affected by the ROM trained, and a joint-angle specific strength increase may occur after partial ROM training. Furthermore, alternating sessions between the initial and final ROMs may increase the maximum strength uniformly among different ROMs, reaching values similar to that achieved from partial ROM training.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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